Shanmugam Subramanian

Mechanical Engineer II | Mechanisms, Fixtures & Robotics | Hyderabad, India

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SUMMARY

Mechanical engineer with 4 years of NPD and NPI experience in electromechanical hardware, plus personal robotics builds: a 6-DOF robotic arm with real-time mirroring, a 1:64 multi-stage gearbox and a cam-lever anchor gripper. Built a pneumatic punching fixture (2 operators to 1, yield 60% to 95%) and fixtures that cut rework by 40%. Experienced in mechanism design, hand calculations, 3D printing and FEA.

EXPERIENCE

JMA Wireless — Hyderabad, India
Mechanical Engineer II May 2023 – Present

Mechanism, Fixture & Automation Design

- Developed a **pneumatic pivot punching fixture** cutting a 2-person task to one operator & **raising yield from 60% to 95%**.
- Built RF PCB fixtures, press-pull tools, & alignment jigs: **40% less rework, 75% faster riveting**, 20–50 min saved per unit.
- Designed pole, wall, ceiling, & tri-sector mounting brackets; gravity-assisted design **halved installation manpower**.
- Introduced topology-optimized FDM 3D-printed prototypes, **cutting 60 g per unit** while meeting stiffness requirements.
- Built Excel-based **hand-calculation tools** for drag force, pneumatic crimping force, & plastic snap-joint behavior.

Product Development & NPI

- Owned end-to-end **NPD & NPI** for eight 5G Macro & Distributed Antenna Systems products for the US & Canada markets.
- Released RF PCB, sheet metal, die-cast, CNC, & EDM parts with a $>$**90% FAI pass rate**.
- Supported **contract manufacturing** setup in Chennai through BOM costing, tooling transfer, & operator training.
- Managed **ECRs, FAIs, & SVPRs** in Salesforce Propel **PLM** & SolidWorks **PDM** with the global team.

CAE & Validation

- Performed **harmonic & random vibration analyses** with 80% first-pass correlation to physical tests (IEC 60068-2-6/64).
- Automated **Ansys Mechanical pre-processing** with Python/ACT scripts (meshing profiles, material assignment, midsurfacing, BCs, solver setup), **cutting setup time $\sim$2 hrs per job** & standardizing team models.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Design Engineer (R&D) Jul 2022 – Apr 2023

- Reverse-engineered & locally manufactured an RF waveguide connector, reducing cost by 8× compared to imports.
- Developed two automotive radar antenna chassis and radomes with **IP68 (IEC 60529)** ratings for ADAS applications.
- Enhanced chassis thermal properties by refining fin geometry, reducing **internal temperature by 6°C**.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Project Intern Mar 2022 – Jun 2022

- Updated automotive radar housing designs based on lead engineer feedback & prepared GD&T-compliant drawings.
- Created company-wide central library in SolidWorks with 700+ frequently used fasteners, reducing design lead time.

EDUCATION

Bachelor of Engineering — Mechanical Engineering GPA: 8.21 / 10
Alagappa Chettiar College of Engineering and Technology (Anna University) 2018 – 2022

PROJECTS AND COMPETITIONS

6-DOF Mimicking Robot— July 2025- Nov 2025 Personal project
Engineered a 6-DOF robotic arm with real-time mirroring, trajectory recording, and repeatable playback of arm movements.

Gearbox design — Dec 2025 Personal project
Designed and built a 1:64 multi-stage spur gear gearbox system to power a conveyor platform for scaled vehicle testing.

Hole Anchor Gripper Personal project
Designed a cam-lever expanding anchor gripper that inserts into holes and locks to lift loads.

SAE Bicycle Design Competition — 2020 - 2021 SAE INDIA
Developed a planetary gear-based bicycle drivetrain achieving 15% higher torque than a typical bicycle.

Prodigy Design Competition— May 2021 NIT Trichy
Designed a regenerative adhesive rolling-belt mechanism for water-body waste removal within a 24-hour time frame; delivered CAD models & documentation, securing runner-up position among 50 teams.

SKILLS & SOFTWARE

Design: SolidWorks, AutoCAD, Fusion 360, Inventor, Onshape, GD&T, Tolerance stackup, 3D Printing, **mechanism, gear, gripper & pneumatic fixture design**
Simulation: Ansys Mechanical & Fluent, Finite Element Analysis, hand-calculation tools
Programming & Embedded systems: Python, MATLAB Simulink & Simscape, Arduino, C, C++
Other Tools: Excel, PowerPoint, PLM (Salesforce Propel), SolidWorks PDM
Management & Quality: RCA, Risk Mitigation, Feasibility Analysis, Cross-functional Team Coordination